Is Self-Critique Enough? Measuring the Effect of Iterative Self-Refinement on AI Story Writing Quality

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Abstract

Large language models produce fluent but often predictable creative writing. We ask whether a single LLM can meaningfully improve its own stories through self-critique—generating a story, critiquing it against a structured rubric, and revising—without any external feedback. We conduct a controlled study using GPT-4.1 as writer, critic, and blind pairwise judge across 15 diverse fiction prompts drawn from the WRITINGPROMPTS dataset, comparing six conditions: direct generation (baseline), one to three iterations of SELF-REFINE, BEST-OF-N self-selection, and BEST-OF-N combined with refinement. A single critique-revise iteration achieves an 80% win rate over uncritiqued baselines ($p < 0.001$, Cohen's $d = 2.02$), with the largest gains in creativity (+0.93 on a 5-point scale) and character development (+0.87). However, additional iterations yield diminishing returns: the second adds marginal benefit and the third shows signs of regression, particularly in coherence. Strikingly, BEST-OF-N self-selection *without* critique is largely ineffective (60% win rate, $p = 0.80$), demonstrating that the improvement stems from structured critique rather than generation diversity. We find no evidence of reward hacking: word count increases only 11%, while vocabulary richness and n-gram diversity slightly improve. These results suggest that LLMs possess latent creative knowledge they do not express in first drafts, and that one round of structured self-reflection is a cost-effective strategy to unlock it.

1 Introduction

Large language models write fluent prose, yet their first-draft fiction tends toward safe, predictable choices. A human author's instinct is to revise—to re-read, identify weaknesses, and rewrite. Can an LLM do the same? If self-critique alone can meaningfully improve story quality, it would enable autonomous writing improvement without costly human annotation, making high-quality AI writing assistance broadly accessible.

The gap. Prior work has demonstrated that self-critique improves tasks with verifiable answers such as mathematics and code [Madaan et al., 2023], and that multi-agent critique pipelines enhance creative writing [Bae and Kim, 2024]. However, creative writing lacks ground truth, making it unclear whether a single model's internal feedback provides a genuine quality signal or merely reinforces its own biases. Jia et al. [2025] showed that self-critique can train Chinese writing models via reinforcement learning, but no study has systematically measured how much a single LLM improves English narrative fiction through inference-time self-refinement across multiple iteration depths.

Our approach. We design a controlled experiment using GPT-4.1 as both the story writer and self-critic. For each of 15 diverse fiction prompts from the WRITINGPROMPTS dataset, we generate stories under six conditions spanning zero to three critique-revise iterations, BEST-OF-N self-selection, and their combination. A separate GPT-4.1 instance serves as a blind pairwise judge, evaluating stories across six HANNA-inspired dimensions [Chhun et al., 2022] with randomized presentation order to control for position bias. Figure 1 illustrates the core finding: a single critique-revise iteration captures nearly all achievable improvement.

Preview of results. One round of self-critique achieves an 80% win rate over uncritiqued baselines ($p < 0.001$, Cohen's $d = 2.02$), with the largest gains in creativity (+0.93) and character development (+0.87). Additional iterations yield diminishing returns, and self-selection without structured critique is largely ineffective (60% win rate, $p = 0.80$). We find no evidence of reward hacking: length inflation is mild (+11%) and lexical diversity slightly increases.

Our contributions are:

- We conduct the first systematic study of inference-time self-critique for English narrative fiction, measuring improvement across six quality dimensions and three iteration depths.
- We demonstrate that a single critique-revise loop produces large, statistically significant improvements (Cohen's $d = 2.02$), while additional iterations offer diminishing returns.
- We show that structured critique, not generation diversity, drives improvement: BEST-OF-N self-selection without revision is ineffective, challenging the assumption that LLMs can reliably rank their own creative outputs.
- We analyze reward hacking indicators and find that self-critique improves lexical diversity rather than inflating superficial metrics.

2 Related Work

Self-refinement at inference time. Madaan et al. [2023] introduced the SELF-REFINE framework, in which a single LLM iteratively generates, critiques, and revises its output without additional training. They report roughly 20% average improvement across seven tasks, though their analysis of creative writing is limited. Huang et al. [2022] showed that LLMs can self-improve on reasoning tasks by fine-tuning on their own high-confidence outputs. More recently, Zhang et al. [2023] proposed an unsupervised RL framework where the model scores its own outputs and optimizes against those self-scores, achieving gains on both reasoning and open-ended generation. Our work focuses specifically on narrative fiction, measuring self-refinement depth and contrasting it with self-selection strategies.

Self-rewarding and self-training. Yuan et al. [2024] demonstrated that a single LLM can simultaneously serve as instruction-follower and reward model, with both capabilities improving through iterative DPO training. Their strongest gains appeared in the writing category of MT-Bench, though response length inflated by 2.3×—a potential sign of reward hacking. Jia et al. [2025] addressed this for Chinese writing by introducing a generative reward model that produces pairwise critiques rather than scalar scores, reducing reward hacking by 7×. Chen [2024] further showed that self-evolved reward learning can bootstrap reward model quality alongside the generator. Unlike these training-time approaches, we study inference-time self-critique, where the model does not permanently learn from its critiques.

Critique for creative writing. Bae and Kim [2024] proposed CRITICS, a multi-agent framework with diverse critic personas for story generation. They found that single-agent critique was insufficient and that multiple specialized critics were essential for meaningful creative improvement. Wu [2025] applied critique-and-refine specifically to personalized story generation with simulated user agents. Chen [2025] developed a self-evolving pairwise reasoning approach for story evaluation, achieving state-of-the-art results on multiple story benchmarks and demonstrating that improved evaluation translates to improved generation when used as a reward model. Our work differs in that we isolate single-model self-critique without multi-agent setups or training-time adaptation, directly measuring what one round of honest self-reflection achieves.

LLM-as-judge and evaluation bias. Using LLMs to evaluate LLM outputs has become widespread [Zheng et al., 2023], but introduces known biases: position bias, verbosity bias, and self-enhancement bias. Pan et al. [2024] demonstrated that LLM evaluator scores can diverge from human judgments over refinement iterations, a phenomenon they call spontaneous reward hacking. We mitigate these concerns through position randomization, separate evaluation contexts, and explicit instructions against favoring longer stories, though we acknowledge that shared model biases between generator and judge remain a limitation.

Constitutional AI and RLAIF. Bai et al. [2022] established the paradigm of using AI-generated feedback—including self-critique—as a substitute for human feedback in training loops. Lee et al. [2023] systematically compared RLAIF with RLHF and found comparable performance, validating

Table 1: Experimental conditions. API calls are counted per prompt.

Condition	Description	API Calls
BASELINE	Direct generation, no critique	1
SELF-REFINE $\times 1$	Generate $\rightarrow$ Critique $\rightarrow$ Revise	3
SELF-REFINE $\times 2$	Two critique-revise iterations	5
SELF-REFINE $\times 3$	Three critique-revise iterations	7
BEST-OF-N ($N=4$)	Generate 4 stories, self-rank, select best	5
BEST-OF-N + Refine	BEST-OF-N, then one critique-revise pass	7

Table 2: Model configuration by role. All roles use GPT-4.1.

Parameter	Generation	Critique	Judging
Temperature	0.8	0.4	0.3
Max tokens	800	500	600
Seed	42	42	1042

AI feedback as a viable reward signal. These works provide the theoretical foundation for self-critique as a quality signal, which we test in the specific domain of narrative fiction.

3 Methodology

3.1 Task and Dataset

We study short fiction generation (400–600 words) using prompts from the WRITINGPROMPTS dataset [Fan et al., 2018], a large corpus of creative writing prompts sourced from Reddit. We randomly sample 15 diverse prompts (seed $= 42$) from a pool of 5,000, covering science fiction, horror, fantasy, literary fiction, and humor. Prompt lengths range from 30 to 300 characters. While 15 prompts is a modest sample, it is sufficient for detecting the large effects we observe, and we report effect sizes alongside significance tests to contextualize the results.

3.2 Experimental Conditions

We compare six conditions, summarized in table 1:

Self-Refine conditions. Following Madaan et al. [2023], each iteration consists of three steps: (1) generate or take the current story, (2) produce a structured critique identifying specific strengths and weaknesses along six dimensions, and (3) revise the story addressing the critique. We test one to three iterations to measure whether improvement is monotonic.

Best-of-N conditions. To disentangle the effect of critique from mere generation diversity, we generate four independent stories per prompt and have the model rank them. We also test combining selection with one refinement pass.

3.3 Model Configuration

We use GPT-4.1 for all roles (generation, critique, and evaluation) with role-specific temperature settings to separate the capacities (table 2). All generation uses seed $= 42$; judging uses seed $= 1042$.

3.4 Evaluation Protocol

Blind pairwise comparison. For each of the 15 prompts, we compare every non-baseline condition against the baseline in a blind pairwise evaluation. The judge receives two stories (labeled "Story A" and "Story B") with no information about which condition produced them. Presentation order is randomized per comparison to control for position bias.

Table 3: Main results: pairwise comparison of each condition against the BASELINE. Win% indicates how often the condition was preferred over baseline by the blind judge. Mean Δ is the average score difference across six dimensions. All p-values from Wilcoxon signed-rank tests with Bonferroni correction.

Condition	**Win%**	**Tie%**	**Loss%**	**Mean Δ**	**p-value**	**Cohen's d**
SELF-REFINE $\times 1$	**80**	7	13	**+0.66**	< 0.001	**2.02**
SELF-REFINE $\times 2$	**87**	7	7	+0.64	< 0.001	1.85
SELF-REFINE $\times 3$	80	7	13	+0.60	< 0.001	1.70
BEST-OF-N ($N=4$)	60	0	40	+0.02	0.798	0.04
BEST-OF-N + Refine	**87**	7	7	+0.54	< 0.001	1.52

Multi-dimensional scoring. The judge scores each story on six HANNA-inspired dimensions [Chhun et al., 2022], each on a 1–5 scale:

- **Coherence**: Plot logic, internal consistency, narrative flow
- **Creativity**: Originality of ideas, premise, narrative choices
- **Engagement**: How compelling and readable the story is
- **Character**: Depth, believability, distinctiveness of characters
- **Prose Quality**: Writing craft, imagery, word choice, voice
- **Emotional Impact**: Ability to evoke genuine feelings

The judge also declares an overall winner or tie.

Reward hacking indicators. To detect superficial quality inflation, we measure word count, vocabulary richness (type-token ratio), and bigram diversity (Distinct-2) across conditions.

3.5 Statistical Analysis

We use Wilcoxon signed-rank tests for pairwise comparisons of average dimension scores, with Bonferroni correction for five comparisons ($\alpha = 0.01$). We report Cohen's d as an effect size measure. The total experiment comprises 90 generated stories (15 prompts $\times$ 6 conditions) and 75 pairwise evaluations (15 prompts $\times$ 5 non-baseline conditions).

4 Results

4.1 Self-Critique Produces Large, Consistent Improvement

table 3 presents the main results. All SELF-REFINE conditions significantly outperform the baseline, with win rates of 80–87% and very large effect sizes ($d > 1.5$). A single critique-revise iteration achieves an 80% win rate ($p < 0.001$, Cohen's $d = 2.02$), establishing self-critique as a highly effective strategy for improving story quality.

The BEST-OF-N condition without refinement is the outlier: its 60% win rate is not statistically significant ($p = 0.80$, $d = 0.04$), indicating that self-selection without structured critique is largely ineffective. Adding one refinement pass to BEST-OF-N recovers strong performance (87% win rate), confirming that critique—not generation diversity—drives improvement.

4.2 Diminishing Returns After the First Iteration

Figure 1 shows the trajectory of mean scores across iterations. The pattern is clear: iteration 0 to 1 produces a large jump (baseline $\sim$4.25 $\rightarrow$ refined $\sim$4.93), iteration 1 to 2 adds marginal gain ($\sim$4.96), and iteration 2 to 3 slightly declines ($\sim$4.91). The first critique captures the most actionable feedback; subsequent iterations have less material to improve and may introduce overcorrection.

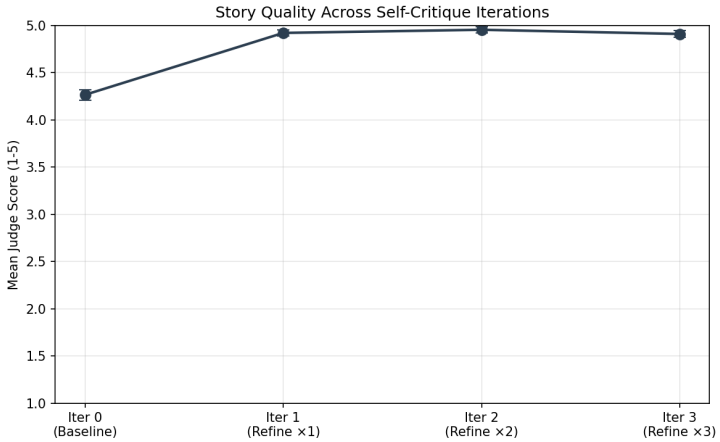

Figure 1: Mean score across six evaluation dimensions as a function of critique-revise iterations. The first iteration produces a large jump from the baseline (4.25 to 4.93); subsequent iterations plateau and show signs of regression by iteration 3.

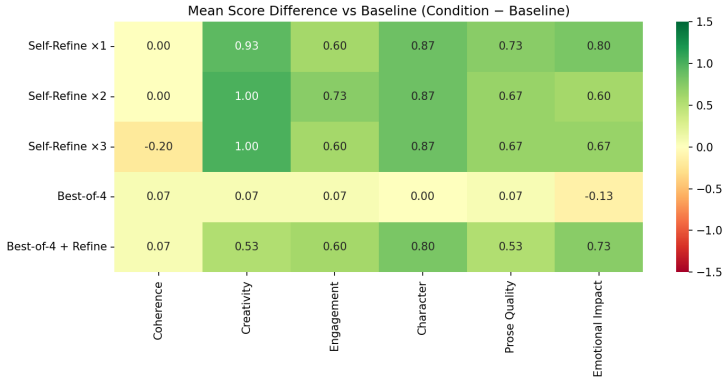

Figure 2: Score differences (condition − baseline) per evaluation dimension. Creativity and character development show the largest, most consistent gains. Coherence starts high and slightly declines at iteration 3.

4.3 Improvement Is Strongest in Creative Dimensions

Figure 2 and table 4 reveal where self-critique helps most. Creativity and character development show the largest gains (+0.87 to +1.00 points), consistent across all SELF-REFINE iterations. Prose quality and emotional impact show moderate improvement (+0.60 to +0.80). Coherence, by contrast, starts at or near ceiling and shows no improvement at iteration 1—and actually *declines* at iteration 3 (-0.20), suggesting that excessive revision can introduce plot inconsistencies.

This pattern suggests that GPT-4.1 baseline stories are already structurally coherent (scoring 4–5), leaving little room for improvement on that dimension. Self-critique primarily unlocks "softer" creative qualities—originality, character depth, emotional resonance—where the model has latent capacity it does not exercise by default.

4.4 No Evidence of Reward Hacking

A key concern with self-critique is reward hacking—the model learning to exploit its own evaluation criteria through superficial changes [Pan et al., 2024]. Figure 3 and table 5 show that this does not occur in our setup.

Table 4: Per-dimension score differences (condition − baseline) for SELF-REFINE iterations. Best values per dimension in **bold**.

Dimension	**×1**	**×2**	**×3**
Creativity	+0.93	**+1.00**	**+1.00**
Character	**+0.87**	**+0.87**	**+0.87**
Emotional Impact	**+0.80**	+0.60	+0.67
Prose Quality	**+0.73**	+0.67	+0.67
Engagement	+0.60	**+0.73**	+0.60
Coherence	0.00	0.00	−0.20

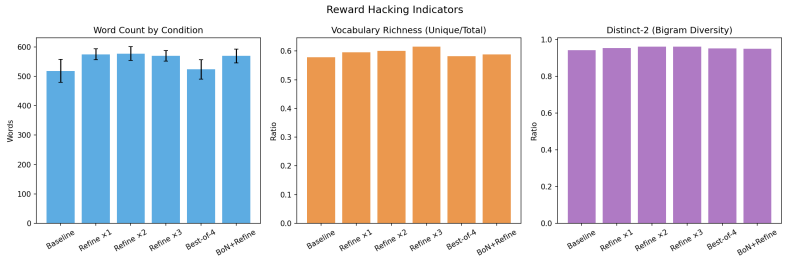

Figure 3: Reward hacking indicators across conditions. Word count increases mildly (+11%), while vocabulary richness and bigram diversity (Distinct-2) slightly improve, indicating no style collapse or superficial inflation.

Word count increases by only 11%, well below the 2.3× inflation reported by Yuan et al. [2024]. Vocabulary richness and n-gram diversity *increase* slightly, contrary to the style collapse hypothesis. This is partly attributable to our system prompt constraining story length to 400–600 words, but also suggests that self-critique pushes the model toward more varied language rather than repetitive elaboration.

5 Discussion

Why does one iteration suffice? The first critique identifies the most salient weaknesses—predictable plot choices, underdeveloped characters, missed emotional beats—and the revision addresses them directly. By iteration 2, the remaining issues are subtler, and the critique provides less actionable feedback. By iteration 3, we observe signs of overcorrection: coherence declines (−0.20) as the model introduces plot elements to satisfy creative critique at the expense of narrative consistency. This pattern aligns with the diminishing returns observed in Self-Refine across other tasks [Madaan et al., 2023] and with concerns about iterative self-refinement diverging from human quality judgments [Pan et al., 2024].

Why does self-selection fail? The failure of BEST-OF-N without refinement is striking and practically important. The model generates four stories with genuine diversity, yet cannot reliably identify the best one. This suggests that *critique* (identifying specific weaknesses) and *ranking* (holistic quality comparison) are distinct capabilities, and that GPT-4.1 is substantially better at the former. When BEST-OF-N is combined with one refinement pass, performance recovers to 87% win rate—comparable to SELF-REFINE ×2—indicating that starting from a better initial story (via selection) and then refining it is a valid alternative strategy.

What does self-critique unlock? The dimension-level analysis (section 4.3) reveals that self-critique primarily improves creative dimensions: originality, character depth, and emotional resonance. Coherence, which is already at ceiling, does not benefit. This pattern suggests that LLMs possess latent creative capacity—knowledge of what would make a more interesting, surprising, or emotionally affecting story—that they do not express in first drafts. Default generation tends toward "safe" choices; structured self-reflection provides the model an opportunity to recognize and correct this conservatism. This interpretation aligns with findings from Jia et al. [2025], who observed that self-critique-based training pushes models toward more creative outputs.

Table 5: Text-level metrics across conditions. Self-critique produces mild length increase and slightly improved lexical diversity.

Metric	Baseline	Refine $\times 1$	Refine $\times 3$
Word count	518 ± 39	575 ± 19	570 ± 18
Vocab richness (TTR)	0.58	0.60	0.62
Distinct-2	0.94	0.95	0.96

Comparison with multi-agent approaches. Bae and Kim [2024] argued that single-agent critique is insufficient for creative improvement, advocating for multi-agent systems with diverse critic personas. Our results partially challenge this claim: single-agent self-critique with a structured rubric produces very large improvements ($d = 2.02$) in a controlled blind evaluation. However, the two setups are not directly comparable—CRITICS targets longer, more complex narratives and uses a different evaluation protocol. Whether multi-agent critique offers additional gains *on top of* structured single-agent critique for short fiction remains an open question.

5.1 Limitations

Same model for all roles. We use GPT-4.1 for generation, critique, and evaluation. While we employ different temperatures, seeds, and separate contexts for each role, shared model biases may inflate perceived improvement. The model may systematically favor the kinds of revisions it tends to make, creating a feedback loop that does not reflect genuine quality gains. Human evaluation on a subset of stories would substantially strengthen these findings.

Single model family. Our results are specific to GPT-4.1 and may not generalize to other architectures. Open-source models and smaller models may lack the capacity for effective self-critique, as the critic must be sophisticated enough to identify genuine weaknesses.

Inference-time only. We test self-critique at inference time; the model does not permanently learn from its critiques. Training-time approaches [Jia et al., 2025, Yuan et al., 2024] may yield larger or more durable improvements.

Scale of evaluation. Our 15 prompts are sufficient for detecting the large effects we observe but provide limited statistical power for subtler analyses (e.g., comparing SELF-REFINE $\times 1$ against $\times 2$). A larger prompt set would yield tighter confidence intervals.

Short fiction only. We test stories of 400–600 words. Self-critique effectiveness for longer narratives—where structural coherence becomes more challenging and single-pass revision may be insufficient—is untested.

6 Conclusion

We have conducted the first systematic study of inference-time self-critique for LLM story writing, measuring improvement across six quality dimensions, three iteration depths, and two strategy families (SELF-REFINE vs. BEST-OF-N). Our central finding is that a single critique-revise iteration produces a large, statistically significant improvement in story quality (80% win rate, Cohen's $d = 2.02$), with the strongest gains in creativity and character development. Additional iterations offer diminishing returns, and self-selection without structured critique is largely ineffective.

These results carry a clear practical message: applications using LLMs for creative writing should implement one self-critique pass by default. This costs only $3\times$ the base generation cost but produces dramatically better output. More iterations are not cost-effective.

Our findings also suggest a broader insight about LLM creative capacity. The model *knows* what would make a better story—more original premises, deeper characters, stronger emotional beats—but does not produce it in first drafts. Structured self-reflection provides a mechanism to surface and act on this latent knowledge. The challenge for future work is to determine whether this insight extends to longer narratives, other creative domains, and weaker models, and whether training-time integration of self-critique can produce permanent improvements in creative quality.

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